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BSIT 301I

Implementation Technology of Enterprise Architecture

In order to align technology (cloud, AI, IoT, cybersecurity) with business strategy and create a comprehensive plan for digital transformation, process optimization, risk reduction, and improved IT governance, Enterprise Architecture (EA) implementation requires strong leadership and organizational buy-in. The implementation technology of Enterprise Architecture is considered a solution to challenges in systems integration, interoperability, and innovation. A study highlighted the potential of IT Enterprise Architecture to systematically unify and optimize organizational processes, improve collaboration, and facilitate information flow. The identified challenges include a significant skill gap among IT professionals, emphasizing the need for development programs and a review of recruitment and training policies to increase the effectiveness of Enterprise Architecture.

<https://www.scitepress.org/Papers/2025/132045/132045.pdf>

<https://www.wwt.com/article/implement-enterprise-architecture>

<https://ieeexplore.ieee.org/document/9141169>

Frameworks Underlying Enterprise Architecture

The concepts and procedures for developing and utilizing an organization's architecture description are outlined in an enterprise architecture (EA) framework. By segmenting the enterprise into domains, layers, or perspectives, it organizes an architect's thoughts and provides models to record those viewpoints for systemic design choices.

Several primary frameworks underlie modern enterprise architecture:

Zachman Framework

It classifies architectural artifacts using a 6x6 matrix and is frequently described as an ontology rather than a methodology. From high-level executive summaries to intricate technical setups, it focuses on six fundamental interrogatives (What, How, Where, Who, When, Why) across six viewpoints.

TOGAF (The Open Group Architecture Framework)

Through its Architecture Development Method (ADM), this de facto industry standard offers a methodical approach to EA design and governance. Business, Data, Application, and Technical (Technology) are its four areas of expertise. A modular framework that specifically

supports contemporary trends like cloud-native architectures, microservices, and digital transformation is introduced in the most recent edition, TOGAF 10.

DoDAF (Department of Defense Architecture Framework)

A data-centric framework that places a high priority on gathering and preserving the information required for wise decision-making. In order to isolate crucial information from underlying complexity, it groups models into multiple perspectives, such as Capability, Operational, Project, and Services.

FEAF (Federal Enterprise Architecture Framework)

Intended for the United States. The federal government uses a Consolidated Reference Model (CRM) to give agencies a uniform language. Strategy, Business, Data, Applications, Infrastructure, and Security are the six domains covered by the CRM.

CRACI (Cloud-native Reference Architecture for the Industrial Compute Continuum)

A contemporary substitute for rigid, hierarchical industrial standards such as RAMI 4.0 or ISA-95. Flexible data flows across the edge-to-cloud continuum are made possible by its decoupled, event-driven approach, which is based on a Digital Twin repository.

These frameworks generally consist of three main areas: architecture descriptions (viewpoints), design methods (processes), and organizational guidance (team structure and governance).

https://en.wikipedia.org/wiki/Zachman_Framework

https://en.wikipedia.org/wiki/The_Open_Group_Architecture_Framework

<https://blogs.bmc.com/togaf-vs-zachman/?print-posts=pdf>

<https://www.cs.unibo.it/cianca/wwwpages/as/4.pdf>

<https://roshancloudarchitect.me/key-differences-between-togaf-9-2-and-togaf-10-f5aa7087578f>

https://obamawhitehouse.archives.gov/sites/default/files/omb/assets/egov_docs/fea_v2.pdf

https://en.wikipedia.org/wiki/Enterprise_architecture_framework

https://dodcio.defense.gov/portals/0/documents/dodaf/dodaf_v2-02_web.pdf

<https://arxiv.org/pdf/1806.03168>

<https://arxiv.org/pdf/2509.07498>

Architecture Components and Artifacts

Building blocks, artifacts, and deliverables are the three basic categories into which architectural work products are organized. These are all controlled in a centralized Architecture Repository that operates as the organization's "single source of truth." The reusable "elements" or components of enterprise capacity are represented by building blocks, which are further separated into Solution Building Blocks (SBBs), which describe the actual implementation components, and Architecture Building Blocks (ABBs), which specify necessary capabilities. Artifacts, which can be classified as catalogs (lists of items), matrices (connection tables), or visual representations that serve as a bridge between technical concepts and stakeholders by lowering complexity, provide more detailed descriptions of particular architectural features. Deliverables are formal, contractually stated outputs, such as an Architecture Definition Document, that are examined and approved by stakeholders at project milestones. These artifacts are usually included in deliverables. Although all deliverables include artifacts, not all artifacts are deliverables because some are made exclusively for internal use. Nevertheless, when taken as a whole, they offer the traceability, consistency, and strategic alignment required to connect business objectives to the underlying technical landscape.

<https://www.linkedin.com/pulse/architectural-content-deliverables-artifacts-building-manish-pandey-y5ruc#:~:text=An%20artifact%20is%20an%20architectural%20work%20product,standard%2C%20or%20snapshot%20of%20the%20Architecture%20Landscape.>

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<https://www.leanix.net/en/blog/enterprise-architecture-artifacts#:~:text=Enterprise%20architecture%20artifacts%20are%20powerful,ensure%20your%20documentation%20is%20perfect.>

<https://haluan.medium.com/maximizing-architectural-artifacts-strategies-for-startups-agile-environments-and-mvp-evolution-fb8eb79971d9>

https://www.youtube.com/watch?v=a_zSn0rn5Ew